

Hierarchical Reinforcement Learning MDP Formulation Underwater Pursuit–Evasion (2D Dynamics)

1 Notation

Pursuer position $\mathbf{p}_t^P = [x_t^P, y_t^P]^\top$, velocity $\mathbf{v}_t^P$, acceleration $\mathbf{a}_t^P$.
 Evader position $\mathbf{p}_t^E$, velocity $\mathbf{v}_t^E$, heading ψ_t^E .
 Relative position $\mathbf{r}_t = \mathbf{p}_t^E - \mathbf{p}_t^P$, $d_t = \|\mathbf{r}_t\|$, $\hat{\mathbf{r}}_t = \mathbf{r}_t/(d_t + 10^{-8})$.
 Relative velocity $\Delta\mathbf{v}_t = \mathbf{v}_t^E - \mathbf{v}_t^P$, $v_t^{\text{close}} = -\Delta\mathbf{v}_t^\top \hat{\mathbf{r}}_t$.
 World bound $[-W, W]^2$, $W = 10$. Catch radius $r_c = 0.5$.
 Time step $\Delta t = 0.05$, $T_{\max} = 500$. $\gamma = 0.99$.

2 Low-level MDP (Environment-level control)

State space $\mathcal{S}_L \subset \mathbb{R}^{22}$ (clipped to $[-10, 10]$):

$$\begin{aligned} \mathbf{s}_t^L = & (x_t^P/W, y_t^P/W, v_{x,t}^P/v_{\max}, v_{y,t}^P/v_{\max}, r_{x,t}/W, r_{y,t}/W, \\ & \Delta v_{x,t}/v_{\max}, \Delta v_{y,t}/v_{\max}, d_t/W, v_t^{\text{close}}/v_{\max}, \hat{v}_{x,t}^E, \hat{v}_{y,t}^E, \\ & \text{dist}_{\text{top}}^P/W, \text{dist}_{\text{bottom}}^P/W, \text{dist}_{\text{left}}^P/W, \text{dist}_{\text{right}}^P/W, \\ & \text{dist}_{\text{boundary}}^E/W, \|\mathbf{v}_t^P\|/v_{\max}, \|\mathbf{v}_t^E\|/v_{\max}, \cos \angle(\mathbf{v}_t^P, \mathbf{r}_t), a_{x,t}^P/a_{\max}, a_{y,t}^P/a_{\max}). \end{aligned}$$

Action space $\mathcal{A}_L = \{0, \dots, 8\}$ (discrete thrust):

- 0: zero thrust; 1...8: full thrust in $\theta_k = (k-1)\frac{\pi}{4}$, thrust $\boldsymbol{\tau} = F_{\max}[\cos \theta_k, \sin \theta_k]$.

Transition P_L : pursuer uses 2D AUV dynamics

$$\begin{aligned} M\dot{\mathbf{v}} &= \boldsymbol{\tau} - D_l\mathbf{v} - D_q\|\mathbf{v}\|\mathbf{v}, \\ \mathbf{a} &= \frac{\boldsymbol{\tau} + \mathbf{F}_{\text{damp}}}{M}, \quad \|\mathbf{a}\| \leq a_{\max}, \quad \|\mathbf{v}\| \leq v_{\max}, \\ \mathbf{v}_{t+1} &= \text{clip}(\mathbf{v}_t + \mathbf{a}\Delta t, v_{\max}), \\ \mathbf{p}_{t+1} &= \mathbf{p}_t + \frac{1}{2}(\mathbf{v}_t + \mathbf{v}_{t+1})\Delta t, \end{aligned}$$

with RK4 integration and boundary reflection (velocity scaled by 0.3 when clamped).

Evader policy (stochastic exogenous) produces desired accel $\mathbf{a}_t^E$ via scripted modes (random/simple/smart/expensive) then obeys turn-rate limit $|\Delta\psi| \leq 0.1$ rad and same damped dynamics with $(v_{\max}^E, a_{\max}^E)$ scaled by difficulty.

Reward $R_L(\mathbf{s}_t^L, a_t, \mathbf{s}_{t+1}^L) = \sum_i r_i$:

$$r_{\text{improve}} = w_1(d_{t-1} - d_t), \quad w_1 = 5.0; \quad r_{\text{close}} = w_2 \text{clip}(v_t^{\text{close}}, -1, 1), \quad w_2 = 1.0;$$

$$r_{\text{align}} = w_3 \cos \angle(\mathbf{v}_t^P, \mathbf{r}_t) \mathbb{1}_{\|\mathbf{v}_t^P\| > 0.1}, \quad w_3 = 0.5;$$

$$r_{\text{trap}} = w_4 \frac{2 - \text{dist}_{\text{boundary}}^E}{2} \mathbb{1}_{\text{dist}_{\text{boundary}}^E < 2}, \quad w_4 = 0.3;$$

$$r_{\text{time}} = -c_t, \quad c_t = 0.05; \quad r_{\text{wall}} = -c_b \mathbb{1}_{\text{hit boundary}}, \quad c_b = 0.1;$$

$$r_{\text{prox}} = 0.2(2 - d_t) \mathbb{1}_{d_t < 2}; \quad r_{\text{catch}} = w_c \mathbb{1}_{d_t < r_c}, \quad w_c = 100.$$

Termination: $d_t < r_c$ (catch, terminal) or $t \geq T_{\text{max}}$ (truncation). Discount $\gamma = 0.99$.

3 High-level MDP (Macro-action planner)

State $\mathcal{S}_H \subset \mathbb{R}^6$:

$$\mathbf{s}_t^H = (\mathbf{r}_t/W, \mathbf{v}_t^P/2, \mathbf{v}_t^E/2).$$

Action $\mathcal{A}_H = \{0, \dots, 8\}$ mapped to waypoint $\mathbf{g}_t$:

$$\mathbf{g}_t = \begin{cases} \mathbf{p}_t^E, & a_t = 0, \\ \mathbf{p}_t^P + d_w[\cos \theta_k, \sin \theta_k], & a_t = k, \quad k \in \{1..8\}, \quad d_w = 2.0, \quad \theta_k = (k-1)\frac{\pi}{4}. \end{cases}$$

High-level transition is semi-MDP: each macro-action invokes low-level controller for $K = 5$ steps; cumulative reward $\tilde{r}_t = \sum_{j=0}^{K-1} r_{t+j}$ and termination if environment terminates within those steps. Effective transition kernel P_H is induced by P_L and the fixed low-level policy π_L (pretrained, frozen).

4 Option-style joint MDP

Define option $o_t \in \mathcal{A}_H$ with duration K ; low-level policy $\pi_L(\cdot | \mathbf{s}^L, o_t)$ is fixed. The induced SMDP tuple $(\mathcal{S}_H, \mathcal{A}_H, P_H, R_H, \gamma)$ has

$$P_H(\mathbf{s}_{t+1}^H | \mathbf{s}_t^H, o_t) = \int P_L^{(K)}(\mathbf{s}_{t+K}^L | \mathbf{s}_t^L, o_t) d\mathbf{s}^L,$$

$$R_H(\mathbf{s}_t^H, o_t) = \mathbb{E} \left[\sum_{j=0}^{K-1} \gamma^j r_{t+j} | \mathbf{s}_t^H, o_t \right].$$

When optimization is separable (bottom frozen), the high-level objective maximizes

$$J_H = \mathbb{E}_{\pi_H, \pi_L} \left[\sum_t \gamma^{tK} R_H(\mathbf{s}_t^H, o_t) \right].$$

5 PPO Objectives

High-level PPO (discrete):

$$L_{\text{actor}} = -\mathbb{E} [\min(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t)],$$

$$L_{\text{critic}} = \frac{1}{2} \mathbb{E} [(V_\phi(\mathbf{s}_t^H) - \hat{R}_t)^2],$$

$$L = L_{\text{actor}} + c_v L_{\text{critic}} - c_e H(\pi_\theta(\cdot | \mathbf{s}_t^H)),$$

with GAE $\hat{A}_t$ and returns $\hat{R}_t = \hat{A}_t + V_\phi(\mathbf{s}_t^H)$. Low-level PPO shares the same form on $(\mathcal{S}_L, \mathcal{A}_L)$.

6 Curriculum and Asymmetry

Difficulty $\delta \in [0, 1]$ rescales evader capability: $v_{\max, \text{eff}}^E = \max(v_{\max}^E \alpha, v_{\max}^P + 0.05)$, $a_{\max, \text{eff}}^E = \min(a_{\max}^E \alpha, a_{\max}^P - 0.05)$ with $\alpha = 0.5 + 0.5\delta$. Initial distance sampled uniformly from $[d_{\min}, d_{\max}]$ per preset (easy/medium/hard/expert). Evader policy switches among scripted modes; thus P_L is Markov but stochastic via exogenous action generation.

7 Summary of MDP Tuples

Low-level $\langle \mathcal{S}_L, \mathcal{A}_L, P_L, R_L, \gamma, T_{\max} \rangle$ with continuous state, 9 discrete actions, deterministic dynamics plus stochastic evader drive, rich shaped reward, termination on catch or horizon.

High-level $\langle \mathcal{S}_H, \mathcal{A}_H, P_H, R_H, \gamma, K \rangle$ SMDP with macro-actions (waypoints), horizon in macro-steps $T_{\max}/K$, reward aggregated from low-level rollouts.